

# WASP-80b

R-2363

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-80\_250730 · created 2026-08-19T23:42:46+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460886.7205 +/- 0.0058 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.23 +/- 0.067                  |
| Transit depth (Rp/Rs)^2                           | 5.31 +/- 3.09 [%]               |
| Orbital Inclination (inc)                         | 87.94 +/- 1.66                  |
| Airmass coefficient 1 (a1)                        | 0.49 +/- 0.22                   |
| Airmass coefficient 2 (a2)                        | 0.64 +/- 0.42                   |
| Scatter in the residuals of the lightcurve fit is | 45.98 %                         |
| Best Comparison Star                              | #3 - [590, 82]                  |
| Optimal Aperture                                  | 0.75                            |
| Optimal Annulus                                   | 3.0                             |
| Transit Duration (day)                            | 0.072 +/- 0.015                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                          |
|---------------------------|------------------------------------------|
| Rp/R■ fit vs published    | 0.23 ± 0.067 vs 0.1714 (deviation 0.61σ) |
| Duration fit vs published | 0.072 d vs 0.0892 d (deviation 0.79σ)    |
| Mid-time O–C              | -6.4 min                                 |
| Depth SNR                 | 2.4                                      |
| Residual scatter          | 45.98 %                                  |

### Light curve

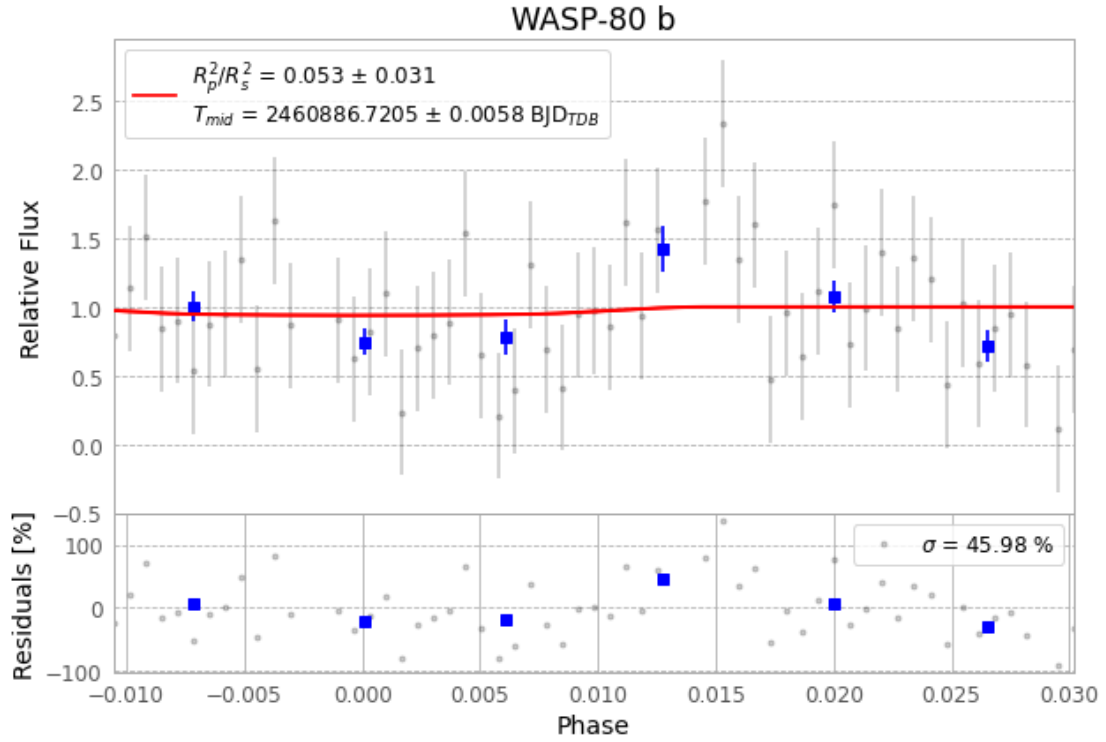


Figure 1 — FinalLightCurve\_WASP-80 b\_2025-07-29.png (SHA-256 349d178041439075...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [524, 117], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 4ad41b4b5b0281b4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

61 FITS frames (40 MB), WASP-80250730042115.FITS → WASP-80250730072115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-80-250730.log (bf03f9fea105...), FinalLightCurve\_WASP-80 b\_2025-07-29.png (349d17804143...), FinalLightCurve\_WASP-80 b\_2025-07-29.csv (ca7c6f1ff49f...)

## Compute resources

Wall time 605.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 23:42Z.